

Classical Mechanics -- Lagrangian Dynamics

Subject: Classical Mechanics | Topic: Principle of Least Action & Euler-Lagrange Equations

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1. Generalized Coordinates & Constraints

Real mechanical systems often subject particles to geometric constraints (holonomic). Using generalized coordinates $q_1, q_2, \dots, q_n$ reduces the equations of motion to the actual degrees of freedom, bypassing unknown constraint forces.

2. Hamilton's Principle of Stationary Action

The actual trajectory followed by a dynamic system between two configurations at times t_1 and t_2 is one that renders the action functional stationary: $\delta S = 0$, where the action S is the time integral of the Lagrangian $L = T - V$.

3. Derivation of Euler-Lagrange Equations

Applying the calculus of variations to $\delta S = 0$ with fixed boundary variations yields: $\frac{d}{dt}(\frac{\partial L}{\partial \dot{q}_j}) - (\frac{\partial L}{\partial q_j}) = 0$ for each coordinate.

4. Conservation Laws and Symmetries

By Noether's theorem, spatial translation symmetry yields linear momentum conservation, rotational symmetry yields angular momentum conservation, and time invariance guarantees conservation of total mechanical energy.